

Chapter 11 Homework: Simultaneous Equations Models

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Exercise 11.24

Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file `fultonfish`.

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/fultonfish.rdata"))
cat("Variable names:", names(fultonfish), "\n")

## Variable names: date lprice quan lquan mon tue wed thu stormy mixed rainy cold tot
r diff change

# Harmonise variable names: dataset may store logs directly as lprice/lquan
if (!"price" %in% names(fultonfish) & "lprice" %in% names(fultonfish))
  fultonfish$price <- exp(fultonfish$lprice)
if (!"quan" %in% names(fultonfish) & "lquan" %in% names(fultonfish))
  fultonfish$quan <- exp(fultonfish$lquan)

head(fultonfish)
```

	date	lprice	quan	lquan	m...	tue	w...	thu	stormy
	<int>	<dbl>	<dbl>	<dbl>	<int>	<int>	<int>	<int>	<int>
1	911202	-0.4307829	8058.003	8.994421	1	0	0	0	1
2	911203	0.0000000	2224.001	7.707063	0	1	0	0	1
3	911204	0.0723207	4231.001	8.350194	0	0	1	0	0
4	911205	0.2471390	5749.998	8.656955	0	0	0	1	1
5	911206	0.6643268	2551.001	7.844241	0	0	0	0	1
6	911209	-0.2065143	10952.000	9.301277	1	0	0	0	0

6 rows | 1-10 of 17 columns

The baseline demand and supply equations are:

Demand: $\ln(QUAN_t) = \alpha_1 + \alpha_2 \ln(PRICE_t) + \alpha_3 MON_t + \alpha_4 TUE_t + \alpha_5 WED_t + \alpha_6 THU_t + e_{1t}$

Supply: $\ln(QUAN_t) = \beta_1 + \beta_2 \ln(PRICE_t) + \beta_3 STORMY_t + e_{2t}$

Part (a): Add MIXED; new reduced-form equation

for $\ln(PRICE)$

```
library(AER)
```

```
# Check column names
names(fultonfish)
```

```
## [1] "date" "lprice" "quan" "lquan" "mon" "tue" "wed" "thu"
## [9] "stormy" "mixed" "rainy" "cold" "totr" "diff" "change" "price"
```

```
# Reduced-form for  $\ln(PRICE)$  with MIXED added to supply side
# (supply instruments: STORMY, MIXED; demand instruments: MON-THU)
rf_model <- lm(log(price) ~ stormy + mixed + mon + tue + wed + thu,
               data = fultonfish)
summary(rf_model)
```

```
##
## Call:
## lm(formula = log(price) ~ stormy + mixed + mon + tue + wed +
##     thu, data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92281 -0.18819  0.00813  0.22404  0.68381
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.35957    0.07915  -4.543 1.50e-05 ***
## stormy       0.44634    0.07921   5.635 1.51e-07 ***
## mixed       0.23688    0.07851   3.017  0.00321 **
## mon        -0.10825    0.10339  -1.047  0.29753
## tue        -0.06609    0.10103  -0.654  0.51446
## wed        -0.04927    0.10377  -0.475  0.63591
## thu         0.03935    0.10071   0.391  0.69681
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3413 on 104 degrees of freedom
## Multiple R-squared:  0.245, Adjusted R-squared:  0.2014
## F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05
```

```
# Test joint significance of STORMY and MIXED
```

```
library(car)
f_supply <- linearHypothesis(rf_model, c("stormy = 0", "mixed = 0"))
print(f_supply)
```

```
##
## Linear hypothesis test:
## stormy = 0
## mixed = 0
##
## Model 1: restricted model
## Model 2: log(price) ~ stormy + mixed + mon + tue + wed + thu
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     106 15.876
## 2     104 12.115  2     3.7604 16.14 7.857e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("STORMY and MIXED jointly significant at 5%",
    f_supply$`Pr(>F)`[2] < 0.05, "\n")
```

```
## STORMY and MIXED jointly significant at 5%? TRUE
```

Part (b): Hausman test for endogeneity of $\ln(PRICE)$

```
# Get reduced-form residuals
vhat <- residuals(rf_model)

# Augmented demand equation
model_hausman <- lm(log(quan) ~ log(price) + mon + tue + wed + thu + vhat,
                    data = fultonfish)
summary(model_hausman)
```

```
##
## Call:
## lm(formula = log(quan) ~ log(price) + mon + tue + wed + thu +
##     vhat, data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1599 -0.3689  0.0807  0.4458  1.2079
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.54023     0.15281  55.889 < 2e-16 ***
## log(price)   -0.93014     0.34483  -2.697  0.00815 **
## mon          -0.01190     0.20332  -0.059  0.95343
## tue          -0.52583     0.19739  -2.664  0.00895 **
## wed          -0.56262     0.20194  -2.786  0.00634 **
## thu           0.09987     0.19792   0.505  0.61491
## vhat          0.48169     0.39473   1.220  0.22511
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6687 on 104 degrees of freedom
## Multiple R-squared:  0.2315, Adjusted R-squared:  0.1872
## F-statistic: 5.221 on 6 and 104 DF, p-value: 9.833e-05
```

```
t_vhat <- coef(summary(model_hausman))["vhat", "t value"]
cat("\nHausman test t-statistic:", round(t_vhat, 4), "\n")
```

```
##
## Hausman test t-statistic: 1.2203
```

```
cat("Conclude endogeneity?", abs(t_vhat) > qt(0.975, df.residual(model_hausman)),
    "\n")
```

```
## Conclude endogeneity? FALSE
```

Part (c): 2SLS demand equation with MIXED added as IV

```
# 2SLS demand equation: instruments include STORMY, MIXED (supply shifters)
model_2sls_demand <- ivreg(log(quan) ~ log(price) + mon + tue + wed + thu |
                           stormy + mixed + mon + tue + wed + thu,
                           data = fultonfish)
summary(model_2sls_demand, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = log(quan) ~ log(price) + mon + tue + wed + thu |
##       stormy + mixed + mon + tue + wed + thu, data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1327 -0.4320  0.0654  0.5133  1.1798
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.54023    0.15661  54.532 < 2e-16 ***
## log(price)  -0.93014    0.35340  -2.632  0.00977 **
## mon          -0.01190    0.20837  -0.057  0.95456
## tue          -0.52583    0.20230  -2.599  0.01068 *
## wed          -0.56262    0.20696  -2.719  0.00767 **
## thu           0.09987    0.20285   0.492  0.62350
##
## Diagnostic tests:
##              df1 df2 statistic  p-value
## Weak instruments    2 104    16.140 7.86e-07 ***
## Wu-Hausman          1 104     1.489   0.225
## Sargan              1  NA     0.772   0.380
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6853 on 105 degrees of freedom
## Multiple R-Squared: 0.185, Adjusted R-squared: 0.1462
## Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317
```

```
cat("\nOLS demand equation for comparison:\n")
```

```
##
## OLS demand equation for comparison:
```

```
model_ols_demand <- lm(log(quan) ~ log(price) + mon + tue + wed + thu,
                        data = fultonfish)
summary(model_ols_demand)
```

```
##
## Call:
## lm(formula = log(quan) ~ log(price) + mon + tue + wed + thu,
##     data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.23844 -0.36738  0.08832  0.42304  1.24866
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.60689     0.14304  60.170 < 2e-16 ***
## log(price)  -0.56255     0.16821  -3.344  0.00114 **
## mon           0.01432     0.20265   0.071  0.94381
## tue          -0.51624     0.19769  -2.611  0.01034 *
## wed          -0.55537     0.20232  -2.745  0.00712 **
## thu           0.08162     0.19782   0.413  0.68073
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6702 on 105 degrees of freedom
## Multiple R-squared:  0.2205, Adjusted R-squared:  0.1834
## F-statistic:  5.94 on 5 and 105 DF,  p-value: 7.08e-05
```

Part (d): Joint significance of MON–THU, STORMY, MIXED in reduced form

```
f_all <- linearHypothesis(rf_model,
                          c("mon = 0", "tue = 0", "wed = 0", "thu = 0", "stormy = 0", "mixed = 0"))
print(f_all)
```

```
##
## Linear hypothesis test:
## mon = 0
## tue = 0
## wed = 0
## thu = 0
## stormy = 0
## mixed = 0
##
## Model 1: restricted model
## Model 2: log(price) ~ stormy + mixed + mon + tue + wed + thu
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     110 16.046
## 2     104 12.115  6     3.9308 5.6238 4.349e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("All jointly significant at 5%?", f_all$`Pr(>F)`[2] < 0.05, "\n")
```

```
## All jointly significant at 5%? TRUE
```

Conclusion: If MON–THU and STORMY/MIXED are jointly significant in the reduced-form equation for $\ln(PRICE)$, the supply equation (with $\ln(PRICE)$ on the right-hand side) is identified.

Part (e): 2SLS supply equation with MON–THU, STORMY, MIXED as IVs

```
# 2SLS supply equation: instruments include MON–THU (demand shifters)
model_2sls_supply <- ivreg(log(quan) ~ log(price) + stormy + mixed |
                           mon + tue + wed + thu + stormy + mixed,
                           data = fultonfish)
summary(model_2sls_supply, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = log(quan) ~ log(price) + stormy + mixed | mon +
##       tue + wed + thu + stormy + mixed, data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.77616 -0.58633  0.02702  0.65208  2.46305
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.3091     0.7166  12.991  <2e-16 ***
## log(price)    1.5171     1.7911   0.847   0.399
## stormy       -1.1121     0.8138  -1.367   0.175
## mixed        -0.5590     0.4765  -1.173   0.243
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments    4 104     0.628  0.6437
## Wu-Hausman         1 106     2.182  0.1426
## Sargan              3  NA     8.244  0.0412 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9688 on 107 degrees of freedom
## Multiple R-Squared:  -0.6596, Adjusted R-squared:  -0.7062
## Wald test: 1.582 on 3 and 107 DF, p-value: 0.1979
```

```
# OLS supply for comparison
model_ols_supply <- lm(log(quan) ~ log(price) + stormy + mixed,
                       data = fultonfish)

cat("\nComparison - Supply Equation:\n")
```

```
##
## Comparison - Supply Equation:
```

```
cat("OLS ln(PRICE) coef:", round(coef(model_ols_supply)["log(price)", 4), "\n")
```

```
## OLS ln(PRICE) coef: -0.4021
```

```
cat("2SLS ln(PRICE) coef:", round(coef(model_2sls_supply)["log(price)", 4), "\n")
```

```
## 2SLS ln(PRICE) coef: 1.5171
```

Exercise 11.28

Supply and demand curves are traditionally drawn with price (P) on the vertical axis and quantity (Q) on the horizontal axis.

The truffle demand and supply equations (from the textbook) are:

Demand: $Q = \alpha_1 + \alpha_2P + \alpha_3PS + \alpha_4DI + e_d$

Supply: $Q = \beta_1 + \beta_2P + \beta_3PF + e_s$

where PS = price of a substitute, DI = per capita disposable income, PF = price of a production factor.

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/truffles.rdata"))
head(truffles)
```

	p <dbl>	q <dbl>	ps <dbl>	di <dbl>	pf <dbl>
1	29.64	19.89	19.97	2.103	10.52
2	40.23	13.04	18.04	2.043	19.67
3	34.71	19.61	22.36	1.870	13.74
4	41.43	17.13	20.87	1.525	17.95
5	53.37	22.55	19.79	2.709	13.71
6	38.52	6.37	15.98	2.489	24.95
6 rows					

```
str(truffles)
```

```
## 'data.frame':   30 obs. of  5 variables:
## $ p : num  29.6 40.2 34.7 41.4 53.4 ...
## $ q : num  19.9 13 19.6 17.1 22.6 ...
## $ ps: num  20 18 22.4 20.9 19.8 ...
## $ di: num  2.1 2.04 1.87 1.52 2.71 ...
## $ pf: num  10.5 19.7 13.7 17.9 13.7 ...
## - attr(*, "var.labels")= chr [1:5] "price of premium truffles, $ per ounce" "quantity of truffles traded in a market period, in ounces" "price of choice truffles (a substitute), $ per ounce" "per capita disposable income, in units of $1000 per month" ...
```

```
names(truffles)
```

```
## [1] "p" "q" "ps" "di" "pf"
```

Part (a): Rewrite equations with P on the left

With price on the vertical axis:

Demand (inverted): $P = \frac{1}{\alpha_2}(-\alpha_1 + Q - \alpha_3PS - \alpha_4DI) + \text{error}$

$$P = \delta_1 + \delta_2Q + \delta_3PS + \delta_4DI + u_d$$

Expected signs: $\delta_2 < 0$ (demand downward sloping), $\delta_3 > 0$ (higher substitute price raises own-good price), $\delta_4 > 0$ (higher income raises price).

Supply (inverted): $P = \frac{1}{\beta_2}(-\beta_1 + Q - \beta_3PF) + \text{error}$

$$P = \gamma_1 + \gamma_2Q + \gamma_3PF + u_s$$

Expected signs: $\gamma_2 > 0$ (supply upward sloping), $\gamma_3 > 0$ (higher factor price raises product price).

Part (b): 2SLS estimates with P on left-hand side

```
# In the inverted system:
# Demand: P ~ Q + PS + DI (IV: PF - supply shifter)
# Supply: P ~ Q + PF (IV: PS, DI - demand shifters)

# Inverted demand equation (instrument: PF)
model_inv_demand <- ivreg(p ~ q + ps + di | pf + ps + di,
                          data = truffles)

summary(model_inv_demand)
```

```
##
## Call:
## ivreg(formula = p ~ q + ps + di | pf + ps + di, data = truffles)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -39.661  -6.781   2.410   8.320  20.251
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -11.428     13.592  -0.841  0.40810
## q             -2.671       1.175  -2.273  0.03154 *
## ps             3.461       1.116   3.103  0.00458 **
## di            13.390       2.747   4.875 4.68e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.17 on 26 degrees of freedom
## Multiple R-Squared:  0.5567, Adjusted R-squared:  0.5056
## Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06
```

```
# Inverted supply equation (instruments: PS, DI)
model_inv_supply <- ivreg(p ~ q + pf | ps + di + pf,
                          data = truffles)
summary(model_inv_supply)
```

```
##
## Call:
## ivreg(formula = p ~ q + pf | ps + di + pf, data = truffles)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.7983 -2.3440 -0.6281  2.4350 11.1600
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -58.7982     5.8592 -10.04 1.32e-10 ***
## q             2.9367     0.2158  13.61 1.32e-13 ***
## pf            2.9585     0.1560  18.97 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.399 on 27 degrees of freedom
## Multiple R-Squared:  0.9486, Adjusted R-squared:  0.9448
## Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16
```

```
cat("\n=== Inverted Demand Equation (P ~ Q + PS + DI) ===\n")
```

```
##
## === Inverted Demand Equation (P ~ Q + PS + DI) ===
```

```
print(coef(model_inv_demand))
```

```
## (Intercept)          q          ps          di
## -11.428407    -2.670519    3.461081    13.389921
```

```
cat("\n=== Inverted Supply Equation (P ~ Q + PF) ===\n")
```

```
##
## === Inverted Supply Equation (P ~ Q + PF) ===
```

```
print(coef(model_inv_supply))
```

```
## (Intercept)          q          pf
## -58.798223    2.936711    2.958486
```

Part (c): Price elasticity of demand “at the means”

```
# Price elasticity = dQ/dP * (P/Q) = (1/delta2) * (P_mean/Q_mean)
# Since P = delta1 + delta2*Q + ..., dP/dQ = delta2, so dQ/dP = 1/delta2
delta2 <- coef(model_inv_demand)["q"]
P_mean <- mean(truffles$p)
Q_mean <- mean(truffles$q)

elasticity_demand <- (1 / delta2) * (P_mean / Q_mean)
cat("Price elasticity of demand at means:", round(elasticity_demand, 4), "\n")
```

```
## Price elasticity of demand at means: -1.2725
```

Part (d): Sketch supply and demand with

$$DI^* = 3.5, PF^* = 23, PS^* = 22$$

```

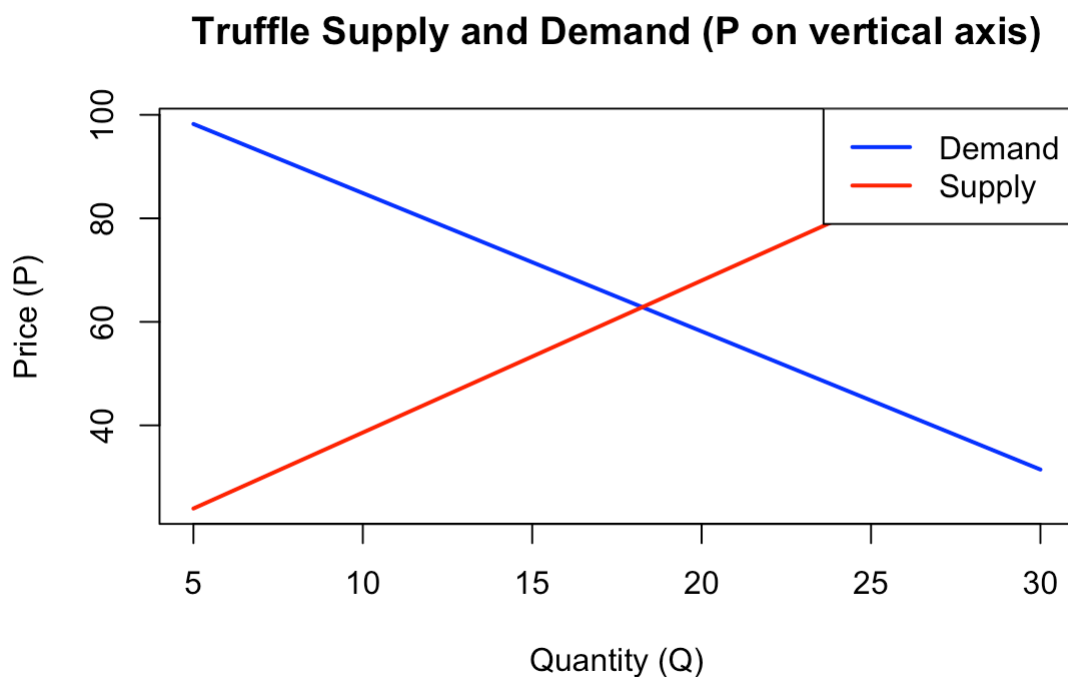
DI_star <- 3.5; PS_star <- 22; PF_star <- 23

# Demand curve: P = delta1 + delta2*Q + delta3*PS + delta4*DI
coef_d <- coef(model_inv_demand)
P_d <- function(Q) coef_d[1] + coef_d["q"]*Q + coef_d["ps"]*PS_star + coef_d["di"]*DI_star

# Supply curve: P = gamma1 + gamma2*Q + gamma3*PF
coef_s <- coef(model_inv_supply)
P_s <- function(Q) coef_s[1] + coef_s["q"]*Q + coef_s["pf"]*PF_star

Q_range <- seq(5, 30, length.out = 200)
plot(Q_range, P_d(Q_range), type = "l", col = "blue", lwd = 2,
     xlab = "Quantity (Q)", ylab = "Price (P)",
     main = "Truffle Supply and Demand (P on vertical axis)",
     ylim = range(c(P_d(Q_range), P_s(Q_range))), na.rm = TRUE))
lines(Q_range, P_s(Q_range), col = "red", lwd = 2)
legend("topright", legend = c("Demand", "Supply"),
     col = c("blue", "red"), lwd = 2)

```



Part (e): Equilibrium P and Q from reduced-form

equations

```
# From the reduced-form estimates (Table 11.2 equivalent)
# Solve:  $P_d(Q) = P_s(Q) \Rightarrow \delta_1 + \delta_2 Q + \delta_3 PS + \delta_4 DI = \gamma_1 + \gamma_2 Q + \gamma_3 PF$ 
#  $(\delta_2 - \gamma_2)Q = \gamma_1 + \gamma_3 PF - \delta_1 - \delta_3 PS - \delta_4 DI$ 
#  $Q^* = (\gamma_1 + \gamma_3 PF - \delta_1 - \delta_3 PS - \delta_4 DI) / (\delta_2 - \gamma_2)$ 

Q_eq <- (coef_s[1] + coef_s["pf"]*PF_star - coef_d[1] -
        coef_d["ps"]*PS_star - coef_d["di"]*DI_star) / (coef_d["q"] - coef_s["q"])
P_eq <- P_d(Q_eq)

cat("Equilibrium Q:", round(Q_eq, 4), "\n")
```

```
## Equilibrium Q: 18.2502
```

```
cat("Equilibrium P:", round(P_eq, 4), "\n")
```

```
## Equilibrium P: 62.8426
```

```
# Also compute from reduced-form equations directly
rf_Q <- lm(q ~ ps + di + pf, data = truffles)
rf_P <- lm(p ~ ps + di + pf, data = truffles)

newdata <- data.frame(ps = PS_star, di = DI_star, pf = PF_star)
Q_rf <- predict(rf_Q, newdata = newdata)
P_rf <- predict(rf_P, newdata = newdata)

cat("\nReduced-form predictions:\n")
```

```
##
## Reduced-form predictions:
```

```
cat("Q* =", round(Q_rf, 4), "\n")
```

```
## Q* = 18.2604
```

```
cat("P* =", round(P_rf, 4), "\n")
```

```
## P* = 62.8154
```

Part (f): OLS estimates (ignoring simultaneity) for

comparison

```
# OLS inverted demand
ols_inv_demand <- lm(p ~ q + ps + di, data = truffles)
# OLS inverted supply
ols_inv_supply <- lm(p ~ q + pf, data = truffles)

cat("=== OLS vs 2SLS Comparison ===\n\n")
```

```
## === OLS vs 2SLS Comparison ===
```

```
cat("---- Demand: coefficient on Q ----\n")
```

```
## ---- Demand: coefficient on Q ----
```

```
cat("OLS: ", round(coef(ols_inv_demand)["q"], 4), "\n")
```

```
## OLS:    0.1512
```

```
cat("2SLS: ", round(coef(model_inv_demand)["q"], 4), "\n")
```

```
## 2SLS:   -2.6705
```

```
cat("\n---- Supply: coefficient on Q ----\n")
```

```
##
## ---- Supply: coefficient on Q ----
```

```
cat("OLS: ", round(coef(ols_inv_supply)["q"], 4), "\n")
```

```
## OLS:    2.6613
```

```
cat("2SLS: ", round(coef(model_inv_supply)["q"], 4), "\n")
```

```
## 2SLS:    2.9367
```

Interpretation: OLS estimates in a simultaneous system are biased and inconsistent because the price and quantity are jointly determined. The 2SLS estimates should be preferred as they account for the simultaneity bias.